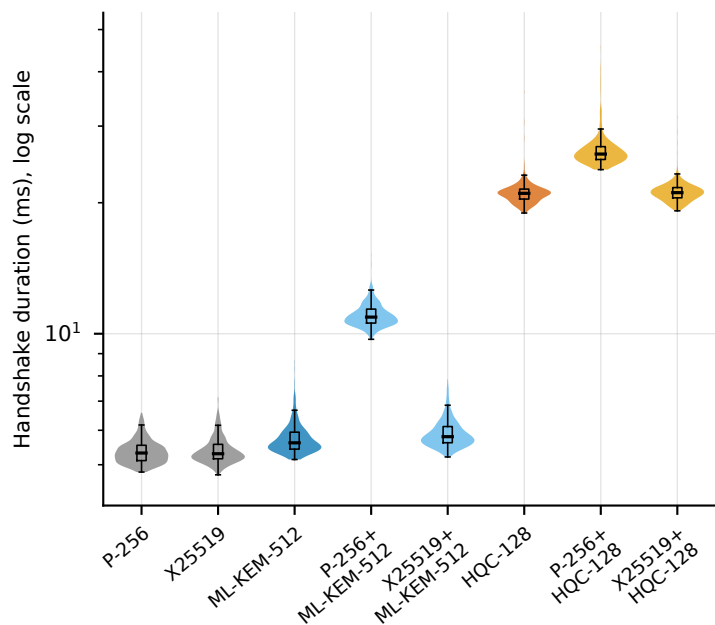


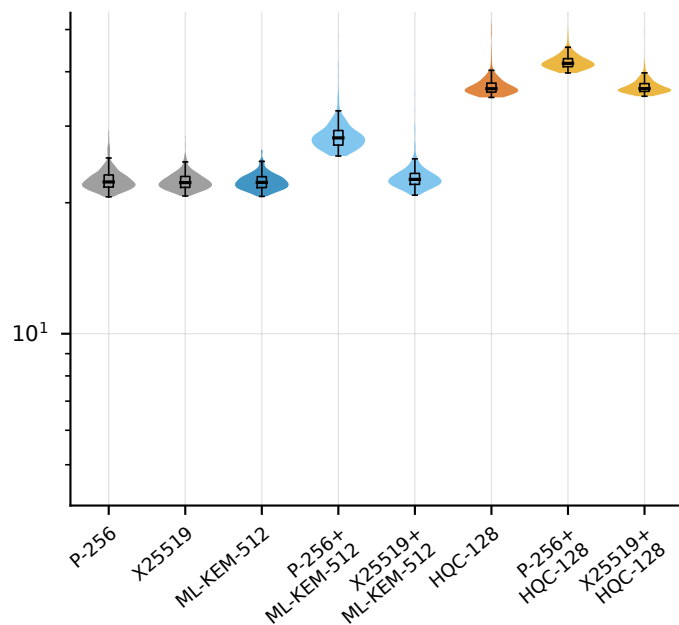
(a)

ed25519 (classical)



(b)

mldsa44 (post-quantum)



(c)

Classical KEX
  ML-KEM (pure)
  ML-KEM (hybrid)
  HQC (pure)
  HQC (hybrid)

| KEM               | ed25519<br>Mean (ms) | ed25519<br>CV (%) | ed25519<br>Max (ms) | mldsa44<br>Mean (ms) | mldsa44<br>CV (%) | mldsa44<br>Max (ms) |
|-------------------|----------------------|-------------------|---------------------|----------------------|-------------------|---------------------|
| P-256             | 5.38                 | 6.4               | 6.58                | 22.6                 | 5.9               | 29.4                |
| X25519            | 5.42                 | 7.1               | 7.15                | 22.4                 | 4.6               | 28.5                |
| ML-KEM-512        | 5.75                 | 8.0               | 8.69                | 22.4                 | 4.4               | 26.5                |
| P-256+ML-KEM-512  | 11.0                 | 6.0               | 15.3                | 28.6                 | 8.0               | 48.7                |
| X25519+ML-KEM-512 | 5.92                 | 7.5               | 7.86                | 22.9                 | 7.0               | 38.6                |
| HQC-128           | 21.0                 | 5.8               | 36.2                | 37.2                 | 7.0               | 58.6                |
| P-256+HQC-128     | 26.4                 | 8.6               | 45.9                | 42.5                 | 7.3               | 71.0                |
| X25519+HQC-128    | 21.2                 | 5.3               | 31.7                | 37.0                 | 4.3               | 55.2                |